

0.0.1 Annulus and spherical-shell Stokes benchmarks

The Cartesian SolCx tests described above provide a controlled verification case for Stokes flow on simple domains. We also tested the Stokes solver in annulus and spherical-shell geometries, where the same finite-element formulation must handle curved boundaries, radial body forces, pressure gauge freedom, and velocity boundary conditions on non-planar surfaces. These tests are closer to the geometries used in mantle-convection models and therefore provide an important complement to the Cartesian benchmarks. The calculations use the Underworld3 finite-element framework [Moresi et al., 2025].

Only volume L_2 -norm errors in velocity and pressure are summarised here. Boundary pressure and stress diagnostics are useful in the standalone benchmark articles, but they are not included in this combined Stokes-solver summary. For a numerical field q_h and analytical reference solution q^* , the relative volume error is

$$E_{L_2}^{\text{rel}}(q) = \left(\frac{\int_{\Omega} |q_h - q^*|^2 d\Omega}{\int_{\Omega} |q^*|^2 d\Omega} \right)^{1/2}. \quad (1)$$

This definition is applied to both the velocity field and pressure field. The pressure is compared after applying a common mean-pressure gauge, because the incompressible Stokes equations determine pressure only up to an additive constant unless an explicit pressure reference is imposed.

Benchmark setup The annulus tests use analytical solutions in polar geometry. The Thieulot–Puckett annulus benchmark is a smooth manufactured Stokes solution with radial gravity and tangential velocity boundary conditions. It is useful for checking the optimal volume convergence expected from mixed finite elements [Thieulot and Puckett, 2018, Girault and Raviart, 1986, Boffi et al., 2013]. The Kramer annulus benchmark separates smooth volumetric forcing from delta-function forcing on an internal circular interface, and includes both free-slip and zero-slip velocity boundary conditions [Kramer et al., 2021].

The spherical-shell tests use analytical solutions in three-dimensional shell geometry. The Thieulot spherical benchmark tests smooth spherical-shell Stokes flow with both constant and radially varying viscosity [Thieulot, 2017]. The Kramer spherical benchmark again separates smooth volumetric forcing from delta-function forcing on an internal spherical interface [Kramer et al., 2021]. This distinction is important because the smooth cases test the polynomial approximation order of the finite element spaces, whereas the delta-function cases intentionally reduce solution regularity.

results For the smooth Thieulot–Puckett annulus benchmark, the Taylor–Hood element pairs recover the expected hierarchy in the volume L_2 norm. The $P_2 \times P_1$ pair gives approximately third-order velocity convergence and second-order pressure convergence. Increasing the polynomial order to $P_3 \times P_2$ improves the observed rates to approximately fourth order in velocity and third order in pressure before the finest meshes approach the error floor. The $P_2 \times P_0$ case gives lower but systematic convergence, with approximately second-order velocity and first-order pressure behaviour.

For the Kramer annulus benchmark, the observed rates depend primarily on the regularity of the forcing. Smooth forcing gives second-order pressure convergence and systematic second-order or better velocity convergence for the $P_2 \times P_1$ discretisation. When the density forcing is concentrated on an internal interface, the solution is less regular and the convergence rates drop to approximately $O(h^{1.5})$ for velocity and $O(h^{0.5})$ for pressure. This reduction is expected from the benchmark construction and is not interpreted as a failure of the Stokes solve.

The spherical-shell Thieulot benchmark gives the cleanest three-dimensional smooth-solution result. For the $P_2 \times P_1$ discretisation, both the constant-viscosity and radially varying viscosity cases

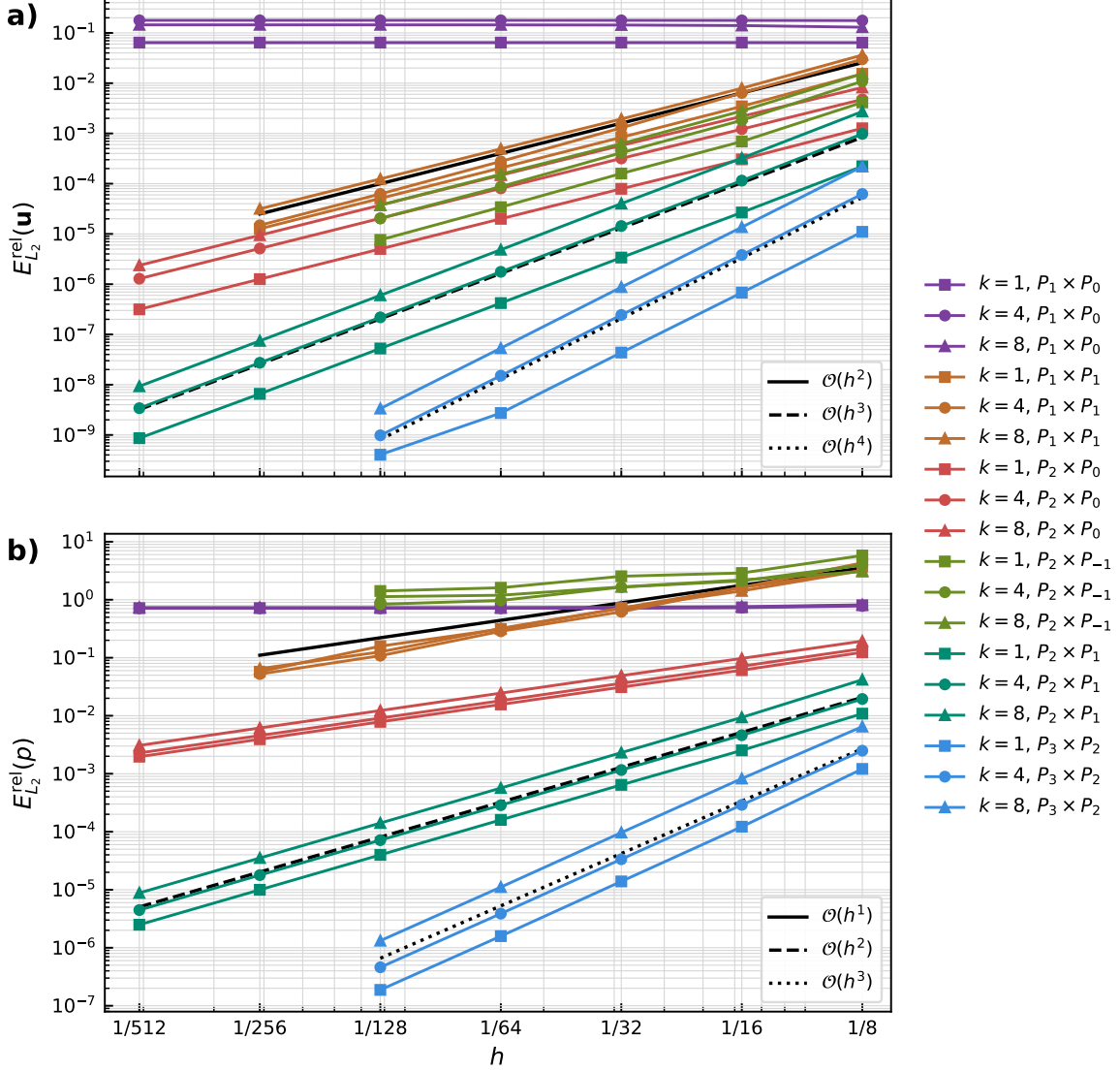


Figure 1: Velocity and pressure convergence for the Thieulot-Puckett annulus benchmark.

approach $O(h^3)$ velocity convergence and $O(h^2)$ pressure convergence in volume norms. This confirms that the spherical-shell Stokes implementation reproduces the expected mixed finite-element convergence hierarchy when the analytical solution is smooth.

The spherical Kramer benchmark shows the same smooth-versus-delta distinction as the annulus Kramer benchmark. Smooth pressure errors converge close to $O(h^2)$, while smooth velocity errors are closer to $O(h^2)$ than to the ideal $P_2 L_2$ rate over the tested mesh sequence. The delta-function cases recover the expected reduced trends of approximately $O(h^{1.5})$ in velocity and $O(h^{0.5})$ in pressure. Taken together, the annulus and spherical-shell benchmarks show that the Stokes implementation reproduces the expected volume-error behaviour for smooth analytical solutions and the expected reduced convergence when the forcing contains an internal singular interface.

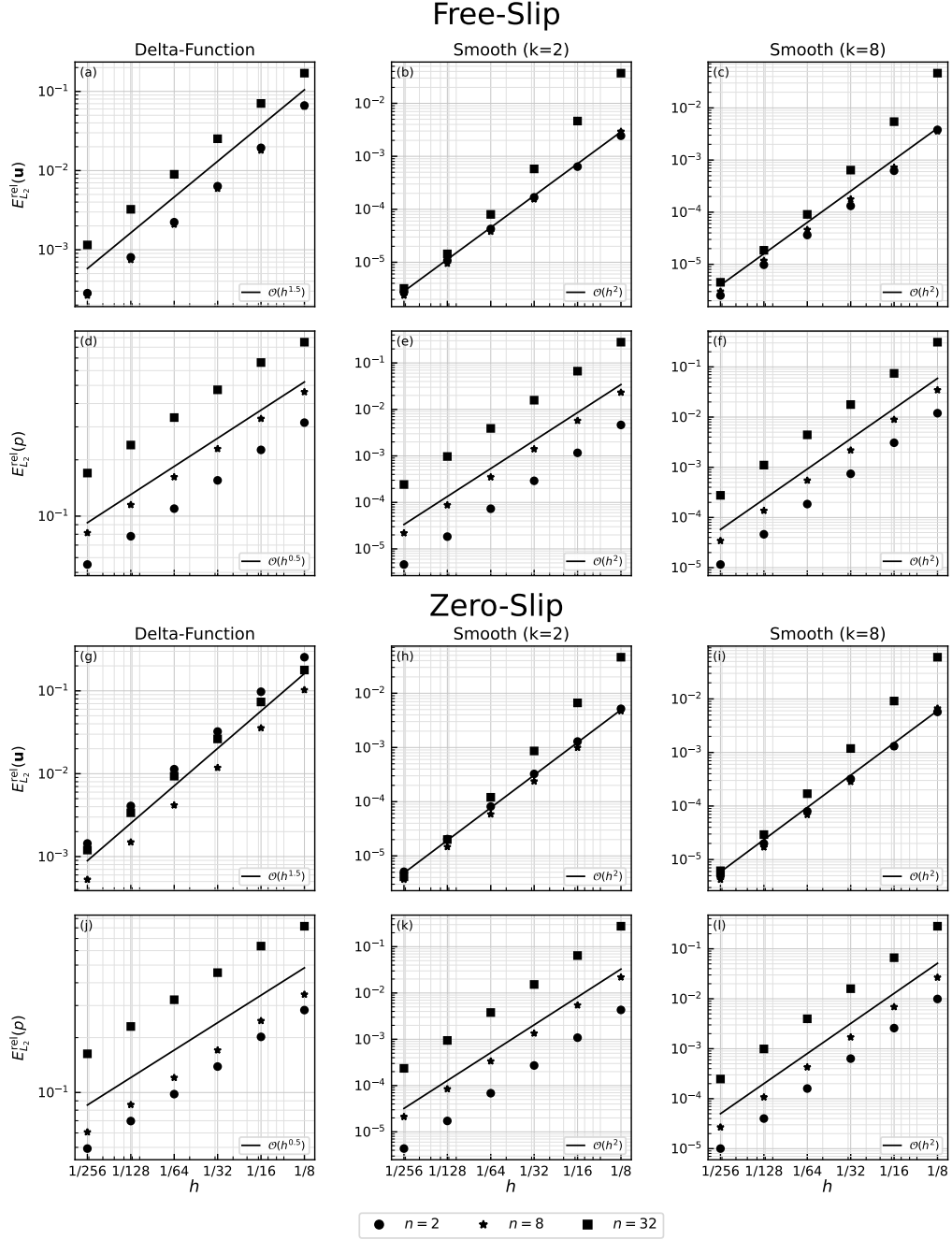


Figure 2: Velocity and pressure convergence for the Kramer annulus benchmark.

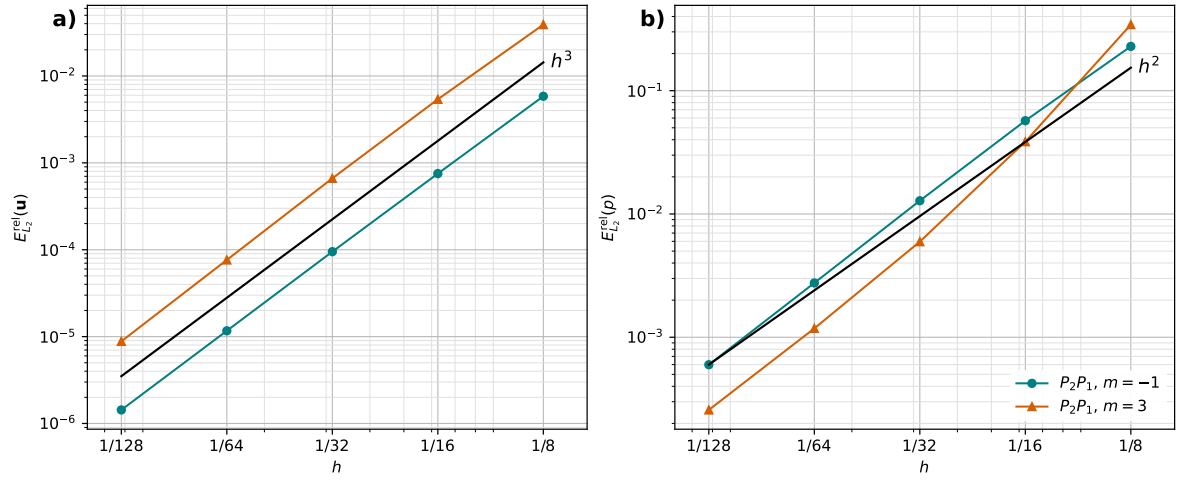


Figure 3: Velocity and pressure convergence for the Thieulot spherical-shell benchmark.

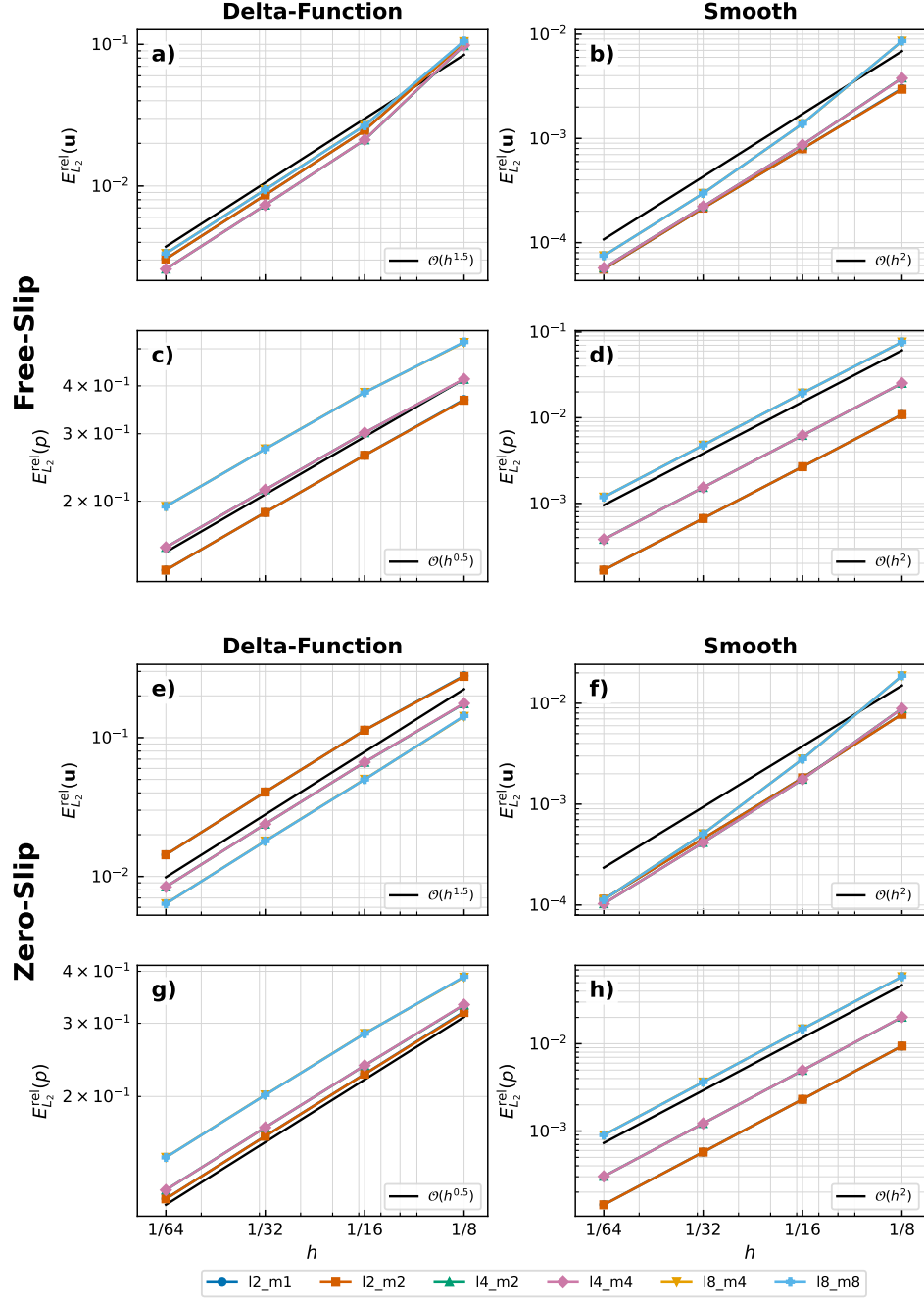


Figure 4: Velocity and pressure convergence for the Kramer spherical-shell benchmark.

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